

EXTREMAL GRAPH THEORY

Erdős Problem 915

Maximising Edge Density Subject to Connectivity Constraints

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The title, in plain words

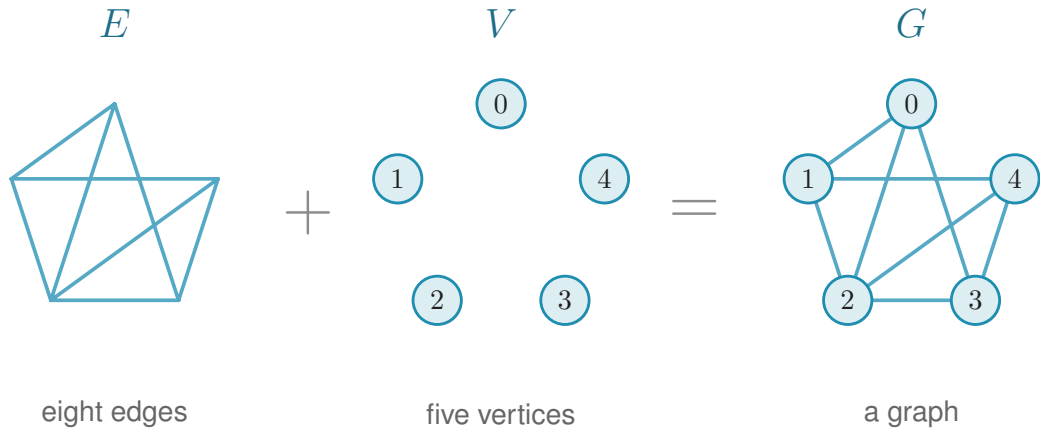
**Maximising edge density in graphs
under connectivity constraints**



translation

networks with many links and low connectivity

A graph is edges and vertices



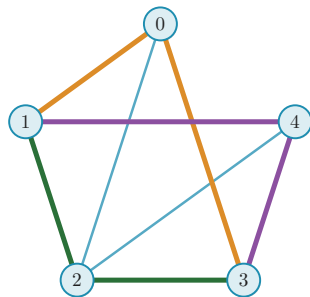
Erdős Problem 915

as many edges as possible, subject to the cap

$$\ell_m(n) = \max\{|E(G)| : \lambda_G^{\max} \leq m-1\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v) \text{ over every pair of vertices}$$

$\lambda_G(u, v)$ = the largest number of $u-v$ routes
that share **no edge**: one colour each



$$\lambda(1, 3) = 3$$

three routes from 1 to 3,
three colours, no edge reused

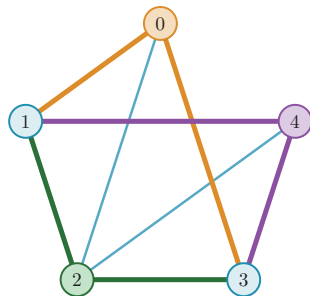
The same question, vertex by vertex

the same cap, read on vertices instead of edges

$$k_m(n) = \max \{ |E(G)| : \kappa_G^{\max} \leq m - 1 \}$$

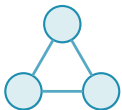
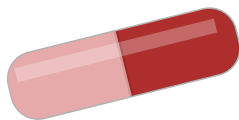
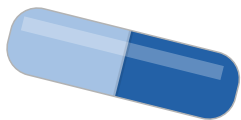
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v) \text{ over every pair of vertices}$$

$\kappa_G(u, v)$ = the largest number of u - v routes that share **no interior vertex**



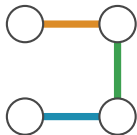
$$\kappa(1, 3) = 3$$

each route keeps its own interior
vertex: 0, 2 and 4



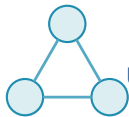
simple

multi



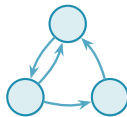
graph

hyper



undirected

directed

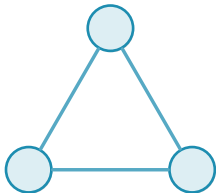


Choice 1 of 3: multiplicity



simple

at most one edge per pair



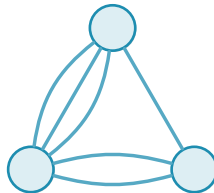
$$\mu(u, v) \in \{0, 1\}$$

VS



multigraph

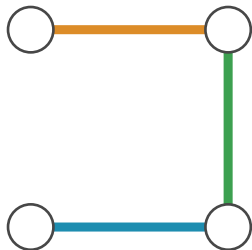
parallel copies allowed



$$\mu(u, v) \in \{0, \dots, m-1\}$$

Choice 2 of 3: arity

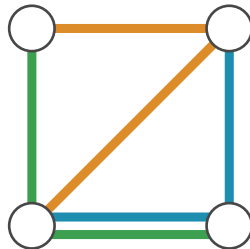
 **graph**
every line serves two stops



$$r = 2$$

VS

 **hypergraph**
a line serves three or more



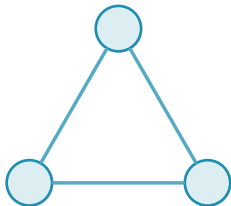
$$r \geq 2, \text{ hyperedges span } r \text{ vertices}$$

Choice 3 of 3: direction



undirected

an edge is an unordered pair



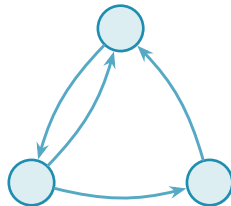
value linear in n

VS



directed

an arc is an ordered pair



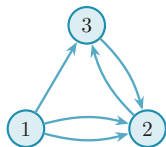
value quadratic in n

PART II

The program

Sixteen variants, one flow computation.

Modelling: two encodings



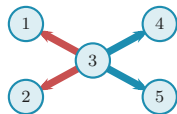
directed multigraph G

$\xrightarrow{\mu}$

$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$

multiplicity matrix μ

$\text{Diag}(\mu) = 0$ (no loops)



directed hypergraph H

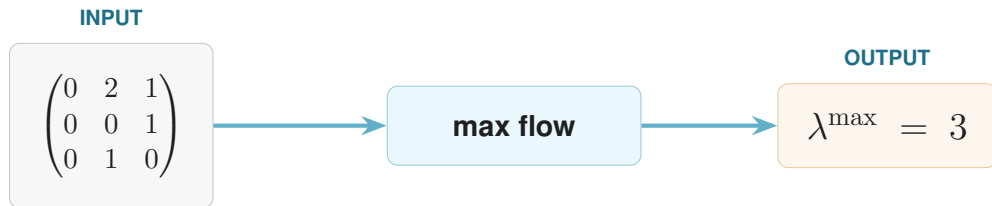
$\longrightarrow$

list of hyperedges:

$e_1 : \{3 \mid 1, 2\}$

$e_2 : \{3 \mid 4, 5\}$

Measuring: one max-flow computation

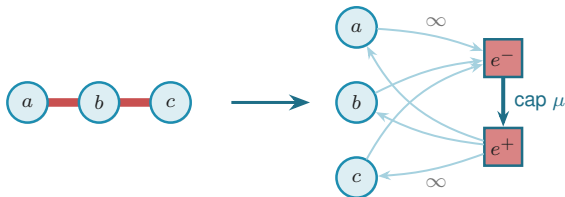


Transformations: everything becomes this one input

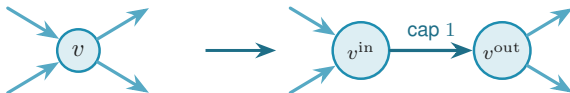
undirected $\rightarrow$
directed



hypergraph $\rightarrow$
graph

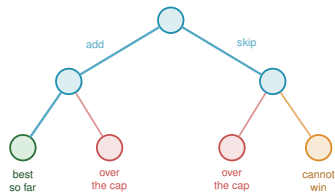


$\kappa \rightarrow \lambda$

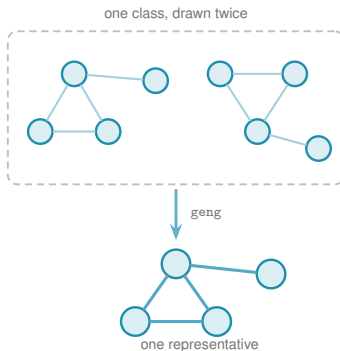


Efficiency: three ways to make the sweep finish

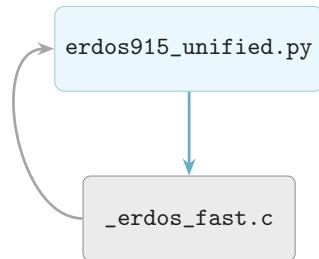
pruning



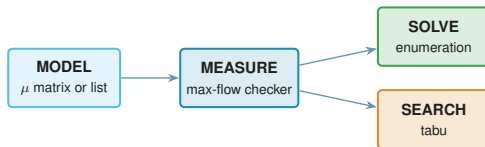
generating search



C: faster max flow



Solving versus searching



solving

- walk every graph on n vertices
- finishes $\Rightarrow$ the exact value
- times out $\Rightarrow$ a lower bound

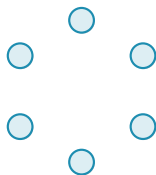
searching

VS

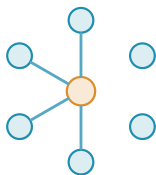
- steer by the energy Φ
- reaches sizes the sweep cannot
- always a lower bound

The search: one number per graph

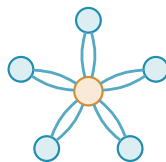
$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \Lambda \text{ large, so breaking the cap never pays}}}$$



$$\Phi = 0$$



$$\Phi = -4$$



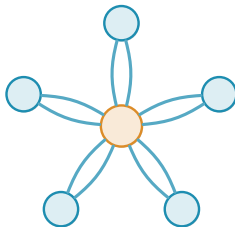
$$\Phi = -10$$

What the search finds on its own



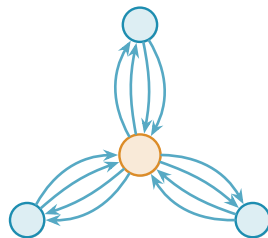
spanning tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

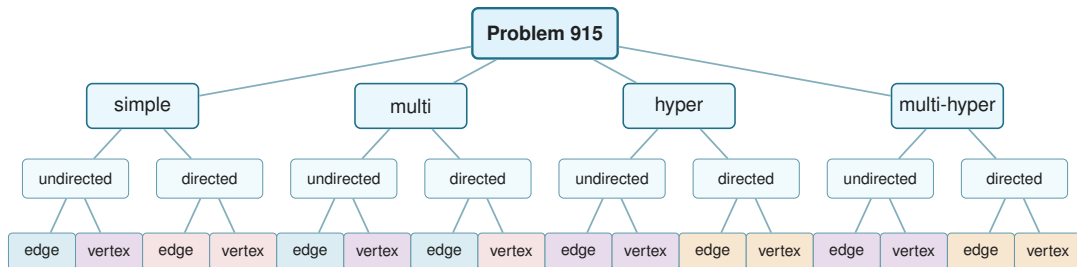
$$n = 6, m = 3, L_3(n) = 10$$



double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

Status of the sixteen variants



closed form proved for every n and m



conjectured, matching certified small n



leading constant proved, exact value open



exact on stated parameter ranges, open outside them

What is new here

1 One-way multigraphs, solved for every n and every m

$L_m^{\text{dir}}(n) = (m-1) \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right)$, by deleting a sparse reachability-preserving subgraph once per unit of $m-1$ rather than a vertex at a time. The quadratic extremisers are classified for $n \geq \max(8, 2m)$.

2 One-way simple graphs: an unconditional quadratic bound

$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1)$, with no hypothesis on the shape of the extremiser, so the leading constant and the order of the next term are theorems. The same count re-runs under the stricter separation.

3 Hyperedge networks: the vertex problem at $m=2$ and $m=3$, and a directed leading term

$\lfloor 2(n-1)/(r-1) \rfloor$ is a universal $m=3$ bound from a new incidence-graph rank lemma, attained for $r \geq 3$ when $2 \leq \binom{n-2}{r-2}$. One-way, the leading term is $(m-1)n^2/(4(r-1))$ under both separations.

What is left

- The undirected *vertex* problem at $m \geq 6$, open since 1974
- The decomposition that would turn the directed conjecture into a theorem: that every extremal non-hub digraph splits as $A \cup B$ with no arc coming back
- Whether the directed vertex value equals the arc value at $m \geq 3$, which Whitney's inequality does not hand over
- The exact values behind the leading terms now settled, one-way and for hyperedges alike

AI: curse or blessing?

Pros

- $\text{\LaTeX}$ typesetting
- length & complexity of generated text

Cons

- length & complexity of generated text
- control

It's a tool.

short leash + AI proofs = argued conjecture [AI]